$$\frac{\mathrm{d}\mathbf{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\,\mathbf{\Gamma}(\boldsymbol{\gamma}(t))$$
$$\mathbf{\Gamma}(\boldsymbol{\gamma}(0)) = \mathbf{\Gamma}(\boldsymbol{\gamma}_0)$$

$$\mathfrak{L} \equiv iL \coloneqq \{\ldots, \mathcal{H}\} \quad \boldsymbol{\gamma} \in X = \mathbb{R}^{6M^{FG}}$$

$$\mathbf{A}(\boldsymbol{\gamma}) = \left\{ A_I(\boldsymbol{\gamma}) \in L^2\left(X, \mathbb{P}\right),\ \text{where}\ \ I = 1, \ldots, M^{CG}\ \ \text{and}\ \ M^{CG} \leq M^{FG} \right\}$$

$$\mathcal{P}_{\mathbf{a}} F(\boldsymbol{\gamma}) = \frac{1}{\Omega(\mathbf{a})} \int_X F(\boldsymbol{\gamma}')\, \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{a}\right) \rho^{eq}(\boldsymbol{\gamma}') \mathrm{d}\boldsymbol{\gamma}', \qquad \Omega(\mathbf{a}) = \int_X \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{a}\right) \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}')$$

$$\mathcal{P}_{\mathbf{a}}^{+} = \mathcal{P}_{\mathbf{a}}$$

$$\mathcal{M}\left(\mathbf{a}, s\right) \coloneqq \beta\, \mathcal{P}_{\mathbf{a}} \left[\left(\mathcal{Q}_{\mathbf{a}} \mathfrak{L}\, \mathbf{\Gamma}_0\right) \otimes \left(\mathbf{e}^{\mathcal{Q}_{\mathbf{a}} \mathfrak{L} s}\, \mathcal{Q}_{\mathbf{a}} \mathfrak{L} \mathbf{\Gamma}_0 \right) \right] \equiv \beta\, \mathbb{E}\left[\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(0, \boldsymbol{\gamma}_0) \otimes \boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(s, \boldsymbol{\gamma}_0) \middle| \mathbf{a} \right]$$

$$\frac{\partial \mathbf{R}_K}{\partial t} = \frac{\mathbf{P}_K}{M_K}$$
$$\frac{\partial \mathbf{P}_K}{\partial t} = -\omega_{N^A K} \frac{\partial U^{AA}}{\partial \mathbf{R}_K} - \frac{\partial V^{\mathrm{eff}}}{\partial \mathbf{R}_K} - \sum_{J=1}^{M^{CG}} \int_0^t \mathcal{M}_{KJ}(t-s, s)\, M_J^{-1}\, \mathbf{P}_J(t-s)\, \mathrm{d}s + \beta^{-1} \sum_{J=1}^{M^{CG}} \int_0^t \frac{\partial \mathcal{M}_{KJ}(s)}{\partial \mathbf{P}_J}\, \mathrm{d}s + \boldsymbol{\mathcal{F}}_K(t)$$

$$-\frac{\partial V^{\mathrm{eff}}}{\partial \mathbf{R}_J} = \mathbb{E}\left[-\sum_{k \in S_J} \frac{M_J}{m_k s_J} \frac{\partial}{\partial \mathbf{r}_k} \left(U^{AB} + U^{BB} \right) \middle| \mathbf{\Gamma} \right]$$

$$\mathcal{M}_{KJ}\left(\mathbf{\Gamma}, t\right) = \beta\, \mathbb{E}\left[\boldsymbol{\mathcal{F}}_K(0) \otimes \boldsymbol{\mathcal{F}}_J(t) \middle| \mathbf{\Gamma} \right]$$

$$\boldsymbol{\mathcal{F}}_K(t) = \mathbf{e}^{t \mathfrak{L}_{\mathcal{Q}}} \mathfrak{L}_{\mathcal{Q}} \mathbf{P}_K$$

$$\frac{\partial \mathbf{R}_K}{\partial t} = \frac{\mathbf{P}_K}{M_K}$$
$$\frac{\partial \mathbf{P}_K}{\partial t} = -\omega_{N^A K} \frac{\partial U^{AA}}{\partial \mathbf{R}_K} - \frac{\partial V^{\mathrm{eff}}}{\partial \mathbf{R}_K} - \sum_{J=1}^{M^{CG}} \widetilde{\mathcal{M}}_{KJ}\, M_J^{-1}\, \mathbf{P}_J + \boldsymbol{\mathcal{F}}_K(t)$$

$$\widetilde{\mathcal{M}}_{KJ}\left(\mathbf{\Gamma}\right) = \frac{1}{\lambda} \mathcal{M}_{KJ}\left(\mathbf{\Gamma}, 0\right) = \frac{1}{\lambda} \beta\, \mathbb{E}\left[\boldsymbol{\mathcal{F}}_K(0) \otimes \boldsymbol{\mathcal{F}}_J(0) \middle| \mathbf{\Gamma} \right]$$

$$\mathcal{H}^X(\mathbf{r}, \mathbf{p}) = \frac{1}{2}\, \mathbf{p}^T \left(M^X\right)^{-1} \mathbf{p}\ + \sum_{i=1}^{M^{FG}-1} U_{i,i+1}^X\left(\mathbf{r}_{i+1} - \mathbf{r}_i\right)\ + U_{M^{FG},1}^X\left(\mathbf{r}_1 - \mathbf{r}_M^{FG} + r^* M^{FG}\right),$$

$$U_{i,i+1}^X(r) = \epsilon_{i,i+1} \left[\left(\frac{r_{i,i+1}^*}{r} \right)^{12} - 2 \left(\frac{r_{i,i+1}^*}{r} \right)^{6} \right], \quad \epsilon_{i,i+1} = \begin{cases} \sqrt{\epsilon_\alpha \epsilon_\beta} & i = 1,\, 2,\, 4,\, 5,\, \ldots,\, M^{FG}-2,\, M^{FG}-1; \\ \epsilon_\alpha\ \ (\text{or } \epsilon_\beta) & i = 3,\, 6,\, \ldots,\, M^{FG}; \end{cases}$$

$$\mathrm{d}R_K = \frac{P_K}{M_K}\, \mathrm{d}t$$
$$\mathrm{d}P_K = -\frac{\partial V^{\mathrm{eff}}}{\partial R_K}\, \mathrm{d}t - \widetilde{\gamma}\, \frac{P_K}{M_K}\, \mathrm{d}t + \left(\frac{2\widetilde{\gamma}}{\beta} \right)^{1/2} \mathrm{d}W_K$$

$$\mathrm{d}R_K = \frac{P_K}{M_K}\, \mathrm{d}t$$
$$\mathrm{d}P_K = -\frac{\partial V^{\mathrm{eff}}}{\partial R_K}\, \mathrm{d}t - \sum_{J=1}^{M^{CG}} \frac{\beta}{2}\, \widetilde{\Sigma}_{KJ}(\widetilde{\mathbf{R}})\, M_J^{-1} P_J\, \mathrm{d}t + \widetilde{\Sigma}_{KJ}(\widetilde{\mathbf{R}})^{1/2}\, \mathrm{d}W_J$$

$$\widetilde{\Sigma}_{KJ}(\widetilde{\mathbf{R}}) = \lambda^{-1}\, \Sigma_{KJ}(\widetilde{\mathbf{R}})$$